

Two-lives probabilities

Exercises

1. For a group of two people aged 60 and 61, we know that the probability that none of them would survive one year more is 0,00003. There are 908725 people aged 60, and 4981 of them die during the next year, which is the probability that a person aged 61 dies during the next year?

Solution:

$$q_{\overline{60:61}} = 0,00003$$

$$\ell_{60} = 908.725$$

$$d_{60} = 4.981$$

$$q_{\overline{60:61}} = q_{60} \cdot q_{61}$$

$$q_{61} = \frac{q_{\overline{60:61}}}{q_{60}} = \frac{0,00003}{\frac{4.981}{908.725}} = 0,0055$$

2. The probability that a person aged 40 reaches 45 is 0,92. Which is the probability that, for a group of two people aged 40, at least one of them reaches 45?

Solution:

$${}_5p_{40} = 0,92$$

$${}_5p_{\overline{40:40}} = 1 - {}_5q_{\overline{40:40}} =$$

$$= 1 - ({}_5q_{40} \cdot {}_5q_{40}) =$$

$$= 1 - [(1 - {}_5p_{40}) \cdot (1 - {}_5p_{40})] =$$

$$= 1 - (1 - 0,92)^2 = 0,9936$$

3. Calculate the probability that a person aged 30 reaches 40 and a person aged 36 reaches 46. We know that the probability that exactly one of them survive 10 years more is 0,16. On the other hand, the last-survivor probability in 10 years is 0,0098.

Solution:

$${}_{10}p_{30:36}^{[1]} = 0,16 \quad {}_{10}q_{30:36} = 0,0098$$

$${}_{10}p_{30:36} + {}_{10}p_{30:36}^{[1]} + {}_{10}q_{30:36} = 1$$

$$\begin{aligned} {}_{10}p_{30:36} &= 1 - {}_{10}p_{30:36}^{[1]} - {}_{10}q_{30:36} = \\ &= 1 - 0,16 - 0,0098 = \\ &= \mathbf{0,8302} \end{aligned}$$

4. We have a group of two people aged 70 and 75, calculate:

- a) Probability that the first death occurs between the sixth and tenth year.
- b) Probability that the first who dies is the one aged 75 and his death occurs between the sixth and tenth year.
- c) Probability that the last death occurs between the sixth and tenth year.
- d) Probability that the last who dies is the one aged 70, and he dies between the sixth and tenth year.

Data:

$$\ell_{70} = 405.864 \quad \ell_{75} = 298.675 \quad \ell_{80} = 179.897 \quad \ell_{85} = 118.979$$

Solution:

$$\begin{aligned} \text{a) } {}_{5/5}q_{70:75} &= {}_5p_{70:75} - {}_{10}p_{70:75} = \\ &= {}_5p_{70} \cdot {}_5p_{75} - {}_{10}p_{70} \cdot {}_{10}p_{75} = \\ &= \frac{\ell_{75}}{\ell_{70}} \cdot \frac{\ell_{80}}{\ell_{75}} - \frac{\ell_{80}}{\ell_{70}} \cdot \frac{\ell_{85}}{\ell_{75}} = \\ &= \mathbf{0,267} \end{aligned}$$

$$\text{b) } {}_{5/5}q_{70:75}^{\frac{1}{2}} = \frac{1}{2} \cdot {}_{5/5}q_{75} \cdot ({}_5p_{70} + {}_{10}p_{70})$$

$${}_{5/5}q_{75} = \frac{\ell_{80} - \ell_{85}}{\ell_{75}} = 0,204$$

$${}_5p_{70} = \frac{\ell_{75}}{\ell_{70}} = 0,74$$

$${}_{10}p_{70} = \frac{\ell_{80}}{\ell_{70}} = 0,44$$

$${}_{5/5}q_{70:75} = \frac{1}{2} \cdot 0,204 \cdot (0,74 + 0,44) = \mathbf{0,12}$$

$$\begin{aligned} \text{c) } {}_{5/5}q_{70:75} &= {}_5p_{70:75} - {}_{10}p_{70:75} = \\ &= (1 - {}_5q_{70:75}) - (1 - {}_{10}q_{70:75}) = \\ &= (1 - {}_5q_{70} \cdot {}_5q_{75}) - (1 - {}_{10}q_{70} \cdot {}_{10}q_{75}) \end{aligned}$$

$${}_5q_{70} = 0,264$$

$${}_5q_{75} = \frac{\ell_{75} - \ell_{80}}{\ell_{75}} = 0,398$$

$${}_{10}q_{70} = 1 - {}_{10}p_{70} = 0,557$$

$${}_{10}q_{75} = \frac{\ell_{75} - \ell_{85}}{\ell_{75}} = 0,602$$

$${}_{5/5}q_{70:75} = \mathbf{0,23}$$

d)

$${}_{5/5}q_{70:75}^2 = {}_{5/5}q_{70} \cdot \left(1 - \frac{1}{2} ({}_5p_{75} + {}_{10}p_{75}) \right) = 0,1463$$

5. Find an expression for calculating the following probabilities for a group of three individuals (x, y, z) :

- Probability that the dissolution (non joint-life probability) of the group is caused by x in n years.
- Probability that in next n years y dies first and afterwards x also dies.

Solution:

$$\begin{aligned}
 {}_nq_{xyz}^1 &= {}_nq_x \cdot {}_np_{yz} + \frac{1}{2} {}_nq_x ({}_nq_y \cdot {}_np_z + {}_np_y \cdot {}_nq_z) + \\
 &\quad + \frac{1}{3} {}_nq_x \cdot {}_nq_y \cdot {}_nq_z \\
 {}_nq_{xyz}^{21} &= \frac{1}{2} {}_nq_x \cdot {}_nq_y \cdot {}_np_z + \frac{1}{6} {}_nq_x \cdot {}_nq_y \cdot {}_nq_z
 \end{aligned}$$

6. The probability that a individual aged 60 survives 10 years more is 0,623 and the non last-survivor probability of a group of two individuals aged 55 in 5 years is 0,992.

Calculate:

- Probability that an individual aged 55 survives 5 years more.
- Probability that exactly one person survives after 5 years from a group to two people aged 55.
- Probability that, from a group fo two people aged 55, at least one of them die between age 60 and 70.

Solution:

a)

$$\begin{aligned}
 {}_5p_{\overline{55:55}} &= 1 - {}_5q_{\overline{55:55}} = 1 - {}_5q_{55} \cdot {}_5q_{55} = 0.992 \\
 {}_5q_{55} &= \sqrt[2]{1 - 0.992} = 0.08944272 \\
 {}_5p_{55} &= 1 - {}_5q_{55} = 1 - 0.08944272 = 0.91055728
 \end{aligned}$$

b)

$$\begin{aligned}
 {}_5p_{\overline{55:55}}^{[1]} &= {}_5p_{55} \cdot {}_5q_{55} + {}_5q_{55} \cdot {}_5p_{55} = 2 \cdot {}_5p_{55} \cdot {}_5q_{55} \\
 &= 2 \cdot (1 - 0.91055728) \cdot 0.91055728 = 0.16288544
 \end{aligned}$$

c)

$$\begin{aligned}
 {}_{5/10}q_{55:55} &= {}_5p_{55:55} - {}_{15}p_{55:55} = ({}_5p_{55})^2 - ({}_5p_{55} \cdot {}_{10}p_{60})^2 \\
 &= 0.91055728^2 - (0.91055728 \cdot 0.623)^2 \\
 &= 0.50731116
 \end{aligned}$$

7. There are two people aged x and y , where $x > y$. Calculate the probabilities ${}_np_x$ and ${}_np_y$ if we know that the last-survivor probability in n years of the group is 0,07 and the probability that exactly one of them survives n years is 0,6.

Solution:

$${}_nq_{\overline{xy}} = {}_nq_x \cdot {}_nq_y = 1 - {}_np_x - {}_np_y + {}_np_{xy} = 0.07$$

$$1 = {}_np_{xy} + {}_np_{\overline{xy}}^{[1]} + {}_nq_{\overline{xy}} = {}_np_{xy} + 0.6 + 0.07$$

$$\left. \begin{array}{l} {}_np_{xy} = 0.33 \\ {}_np_x + {}_np_y = 1.26 \end{array} \right\} \begin{array}{l} {}_np_x = 0.37234 \\ {}_np_y = 0.88865 \end{array}$$

8. Given two individuals aged 44 and 50, the probability that both of them survive after 10 years is 0,799. Calculate:

- a) Probability that an individual aged 50 dies in the tenth year.
b) Probability that an individual aged 44 survives 10 years later.

Data: ${}_{10}p_{50} = 0,871$; ${}_7p_{50} = 0,936$ and ${}_2p_{57} = 0,989$.

Solution:

$$\begin{aligned} \text{a)} \quad {}_{9/1}q_{50} &= {}_9p_{50} \cdot q_{59} = ({}_2p_{57} \cdot {}_7p_{50}) \cdot (1 - p_{59}) \\ &= ({}_2p_{57} \cdot {}_7p_{50}) \cdot \left(1 - \frac{{}_{10}p_{50}}{{}_9p_{50}}\right) \\ &= ({}_2p_{57} \cdot {}_7p_{50}) \cdot \left(1 - \frac{{}_{10}p_{50}}{{}_2p_{57} \cdot {}_7p_{50}}\right) \\ &= 0.989 \cdot 0.936 \cdot \left(1 - \frac{0.871}{0.989 \cdot 0.936}\right) \\ &= 0.054704 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad {}_{10}p_{44:50} &= {}_{10}p_{44} \cdot {}_{10}p_{50} \\ {}_{10}p_{44} &= \frac{{}_{10}p_{44:50}}{{}_{10}p_{50}} = \frac{0.799}{0.871} = 0.917336 \end{aligned}$$

9. For a group of three individuals (x, y, z) , we know that the probability that in 5 years at least two of them survive is 0,869 and the probability that two of them survive as maximum is 0,41. Calculate the probability that exactly two of them survive in 5 years.

Solution:

$$\begin{aligned}
 {}_5p_{\overline{xyz}}^2 &= {}_5p_{\overline{xyz}}^{[2]} + {}_5p_{xyz} \\
 {}_5p_{\overline{xy}z^2} &= {}_5q_{\overline{xyz}} + {}_5p_{\overline{xyz}}^{[1]} + {}_5p_{\overline{xyz}}^{[2]} \\
 {}_5p_{\overline{xyz}}^2 + {}_5p_{\overline{xy}z^2} &= 1 + {}_5p_{\overline{xyz}}^{[2]} \\
 {}_5p_{\overline{xyz}}^{[2]} &= {}_5p_{\overline{xyz}}^2 + {}_5p_{\overline{xy}z^2} - 1 \\
 &= 0.869 + 0.41 - 1 = 0.279
 \end{aligned}$$